Determination of the impurity concentrations in a semiconductor from Hall coefficient measurements

By P. A. Lee, B.Sc., Ph.D., A.Inst.P., The British Thomson-Houston Co. Ltd., Rugby

[*Paper received* 26 *February*, 1957]

The equation is derived for the carrier concentration as a function of temperature and the concentrations of the acceptors and donors in a semiconductor. As a result, it is shown how these impurity concentrations may be found experimentally from a determination of the Hall coefficient as a function of the temperature. These methods are applied to the case of *p*-type silicon which is dealt with in some detail.

Much interest has been shown in recent years in developing methods for analysing the impurity concentration in semiconductors and in particular in germanium and silicon, e.g. radiochemical[1] and mass spectrometer techniques.[2] These semiconductors may now be produced with impurities which are well beyond the limit of detection of most of these methods and it is necessary to resort to the measurement of the Hall coefficient as a means of analysis. Several authors have given the results of their analysis of the impurity concentrations in semiconductors from the measurement of the Hall coefficient as a function of the temperature.[3,4] This technique has been used in this Laboratory for some time for studying semiconductors and it is shown here how this analysis may be most conveniently carried out.

In an extrinsic semiconductor the carrier concentration is a function of both the temperature and also the impurity concentration which may consist of both donors and acceptors. This function is derived here for a *p*-type semiconductor where the acceptor concentration is in excess of the donor concentration, but the calculations apply equally to an *n*-type semiconductor with an obvious interchange of parameters.

SEMICONDUCTOR WITH BOTH DONORS AND ACCEPTORS

We will assume the conventional case where the donor impurity level is near to the conduction band and the acceptor impurity level close to the valency band. We assume furthermore that the donors are fully ionized and thus the concentration of holes p is given by the mass action relation:

$$pN_A^-/N_A^\times = K_A \tag{1}$$

where N_A^- is the number of ionized acceptors per cubic centimetre, $N_A^\times$ the number of neutral acceptors per cubic centimetre, and K_A the equilibrium constant.

From statistics we know that

$$K_A = (2\pi m^* kT/h^2)^{3/2} \exp(-E_A/kT) \tag{2}$$

where m^* is the effective mass of a hole, and E_A the ionization energy of acceptors.

If N_A is the total number of acceptors per cubic centimetre, then

$$N_A = N_A^- + N_A^\times \tag{3}$$

and if N_D is the total number of donors per cubic centimetre, then for neutrality:

$$p = N_A^- - N_D \tag{4}$$

From equations (1), (3) and (4) it follows that:

$$p = \left(\frac{K_A + N_D}{2}\right)\left\{\left[1 + \frac{4K_A(N_A - N_D)}{(K_A + N_D)^2}\right]^{\frac{1}{2}} - 1\right\} \tag{5}$$

This is the equation that we require and holds from very low temperatures up to temperatures where the intrinsic properties of the semiconductor start to predominate over the extrinsic properties. For convenience we may consider two distinct regions.

(i) At high temperatures where all the impurities are ionized we have:

$$K_A \gg N_A \quad \text{and} \quad K_A \gg N_D$$

and therefore it follows that:

$$\frac{4K_A(N_A - N_D)}{(K_A + N_D)^2} \to \frac{4(N_A - N_D)}{K_A} \ll 1$$

By expanding $\left[1 + \frac{4(N_A - N_D)}{K_A}\right]^{\frac{1}{2}}$ into a binomial series we get:

$$p = N_A - N_D \tag{6}$$

In this region the carrier concentration p is independent of the temperature.

(ii) At very low temperatures we have:

$$K_A \ll N_A \quad \text{and} \quad K_A \ll N_D$$

and therefore it follows from equation (5) that:

$$p = [K_A(N_A - N_D)/N_D] \tag{7}$$

Thus at these temperatures p is proportional to

$$\exp(-E_A/kT)$$

from equations (2) and (7), the constant of proportionality depending only on the ratio of the impurities N_A/N_D.

There is an intermediate region towards which the curve will tend only if $N_A \gg N_D$ (or more exactly if $N_A - N_D \gg K_A$ and $K_A \gg N_D$).

In this case $\frac{4K_A(N_A - N_D)}{(K_A + N_D)^2} \gg 1$ and thus:

$$p = [K_A(N_A - N_D)]^{\frac{1}{2}} \tag{8}$$

In this region p is proportional to $\exp(-E_A/2kT)$.

Graphs of equation (5) are shown in Fig. 1 where the hole concentration p is plotted on a log scale along the ordinate and the inverse of the temperature $1/T$ along the abscissa. Three cases are shown.

(i) Fig. 1(*a*) shows the case of $N_A - N_D > N_D$ and refers to a strongly *p*-type sample.

(ii) Fig. 1(*b*) shows the case of $N_A - N_D < N_D$ where the impurities are nearly balanced with the acceptors in excess.

(iii) Fig. 1(*c*) shows the exceptional but unlikely case of $N_A - N_D = N_D$ or $N_A = 2N_D$ exactly.

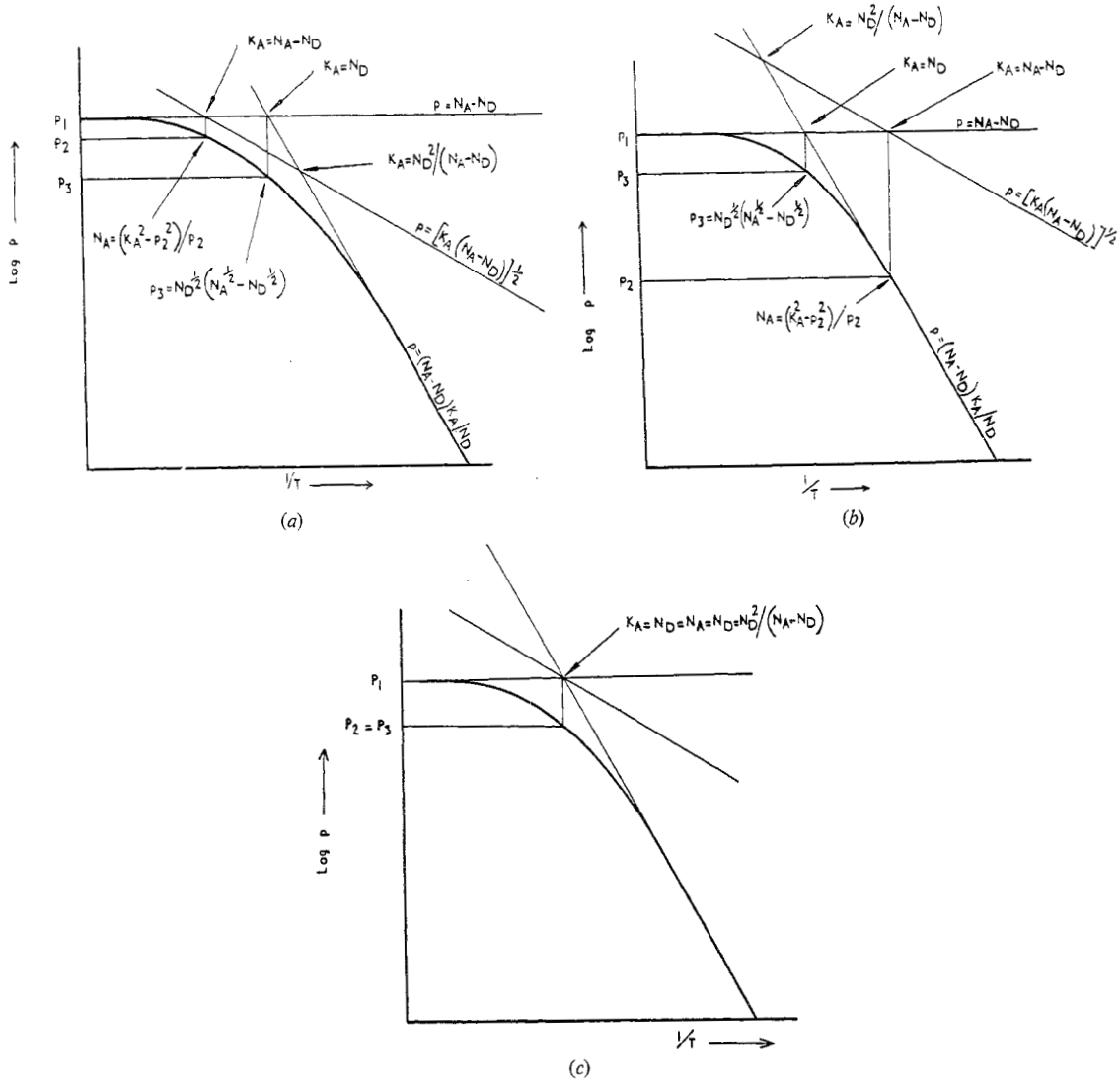


Fig. 1. Graph of carrier concentration *versus* reciprocal temperature
(*a*) $N_A - N_D > N_D$. (*b*) $N_A - N_D < N_D$. (*c*) $N_A - N_D = N_D$.

These criteria form a useful definition when referring to so-called "balanced or unbalanced" semiconductors.

Also shown in Fig. 1 are the straight lines given by equations (6), (7) and (8). Various intercepts which are of interest have been calculated and are as indicated in Fig. 1. It is to be noted that the position of the horizontal line given by equation (6) depends only on $N_A - N_D$ while the intercept of this line with that given by equation (7) depends only on N_D. That is, for a fixed N_A, the intercept will move to lower or higher temperatures according as N_D is decreased or increased, since $N_D = K_A$ at this point.

SEMICONDUCTOR WITH ACCEPTORS ONLY

If we put $N_D = 0$ in equation (5) we obtain:

$$p = \frac{K_A}{2}\left[\left(1 + \frac{4N_A}{K_A}\right)^{\frac{1}{2}} - 1\right] \tag{9}$$

As before we can divide the temperature dependance of p into two distinct regions:

(i) At high temperatures $K_A \ll N_A$ and thus

$$\left.\begin{aligned}\left(1 + \frac{4N_A}{K_A}\right)^{\frac{1}{2}} &\to 1 + \frac{2N_A}{K_A}\\ \text{and} \qquad p &= N_A\end{aligned}\right\} \tag{10}$$

(ii) At low temperatures $K_A \ll N_A$ and thus

$$\left.\begin{aligned}4N_A/K_A &\gg 1\\ \text{and} \qquad p &= (K_A N_A)^{\frac{1}{2}}\end{aligned}\right\} \tag{11}$$

In this range p is proportional to $\exp(-E_A/2kT)$ and the curve does not change its gradient to $\exp(-E_A/kT)$ as is the case if donors are present.

DETERMINATION OF N_A AND N_D FROM THE HALL COEFFICIENT

The Hall coefficient R for a rectangular filament is given by

$$R = \frac{V_H t}{Hi} \times 10^8 \text{ cm}^3 \text{ coulomb}^{-1} \tag{12}$$

where V_H is the Hall voltage in volts, t the thickness of specimen in centimetres, H the magnetic field in gauss, and i the current through specimen in amperes.

In the extrinsic range the carrier concentration p is given by the relation:

$$p = 3\pi/8Re \tag{13}$$

where e is the electronic charge equal to $1{\cdot}6 \times 10^{-19}$ coulombs.

Thus we may obtain an experimental curve of p as a function of the temperature by measuring the Hall coefficient R over a range of temperatures.

To determine the impurity concentration N_A and N_D we must compare equation (5) with this experimental curve. Basically this implies a determination of K_A as a function of temperature, which in turn implies finding the ionization energy of the acceptors E_A and the effective mass of holes m^*. The value of E_A may be found from the curve at low temperatures [equation (7)]. The effective mass may be assumed from other data, e.g. cyclotron resonance[5] or if not otherwise known may be used as an adjustable parameter to obtain a better fit between the experimental and theoretical curves.

An alternative and perhaps simpler method is as follows. At the temperature at which the two tangents to the curve intercept it is seen from Fig. 1 that the value of p is given by:

$$p_3 = N_D^{\frac{1}{2}}(N_A^{\frac{1}{2}} - N_D^{\frac{1}{2}}) \tag{14}$$

If p_1 is the saturation value of p given by equation (6) then it follows that:

$$N_A = [(p_1 - p_3)^2]/(p_1 - 2p_3) \tag{15}$$

$$N_D = p_3^2/(p_1 - 2p_3) \tag{16}$$

It is interesting to note that since:

$$p_1/p_3 = (N_A/N_D)^{\frac{1}{2}} + 1 \tag{17}$$

and $N_A > N_D$ for a p-type semiconductor then

$$p_1 > 2p_3 \tag{18}$$

A certain amount of care is necessary when applying the above procedure for two reasons. Firstly it is to be noted from equations (2) and (7) that it is $\log_e (p/T^{3/2})$ which is proportional to $1/T$ and not $\log_e p$. This means that in some instances the value of p_3 may be too high and it is necessary to correct for this. This is found to be particularly important for balanced samples where it is sometimes found that equation (18) does not hold, but applying the above correction usually overcomes this difficulty. Secondly, if acceptors are present only, then the point p_3 will not exist and p_2 may be used in error. In practice, there are some donors, and measurements at sufficiently low temperatures will reveal any mistake that might otherwise be made.

Should facilities not be available for lowering the temperature to the extent that we are able to draw the tangent given by equation (7) a rough estimate of the impurities may be made using the following procedure. In many cases we know the identity of the majority impurity and merely wish to know its magnitude. In this case we may be able to plot a graph of K_A as a function of temperature.

First of all we obtain $N_A - N_D$ from the value at which p saturates (p_1). We now calculate the temperature at which K_A equals this value and from the curve find the value of $p(p_2)$ at this temperature. From Fig. 1 it follows that:

$$N_A = (K_A^2 - p_2^2)/p_2 \tag{19}$$

In practice this method does not yield very accurate results, and is only mentioned for completeness.

Using this method we can also calculate the limits of determination of impurities in a semiconductor as shown in the next section.

p-TYPE SILICON

It is well known that even after extensive purification of silicon the predominant impurity remaining is boron, which makes the silicon p-type in character. The above analysis was developed to investigate these samples where the majority impurity N_A may be as low as 10^{13} atoms per cubic centimetre. Fig. 2 shows a graph of K_A [equation (2)] plotted on a log scale *versus* the reciprocal of the temperature for the case of boron as an acceptor where it is assumed that $E_A = 0{\cdot}045$ eV[6] and $m^* = 0{\cdot}4$ m[7] where m is the free electron mass.

As indicated above we can find the limits of detection for various temperature ranges as follows for boron as the major impurity from Fig. 2.

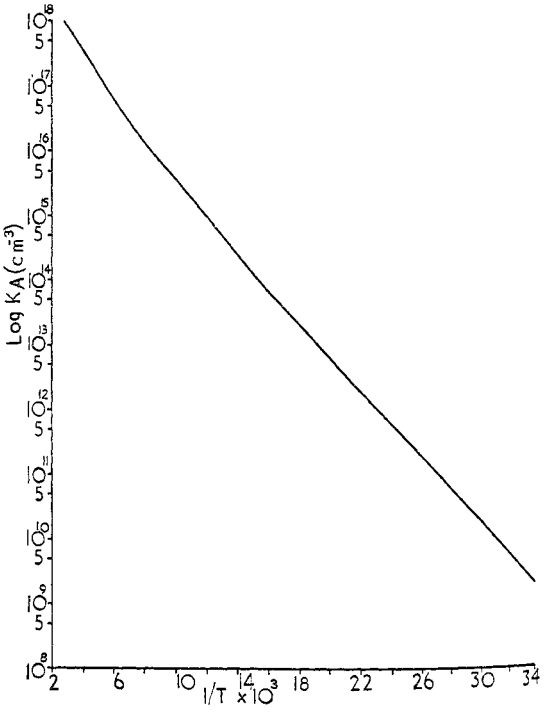


Fig. 2. Graph of the equilibrium constant K_A for $E_A = 0{\cdot}045$ eV and $m^* = 0{\cdot}4$ m

(i) For measurements down to 90° K (boiling point of oxygen), we may find N_A and N_D providing $(N_A - N_D) > 2 \times 10^{15}$ cm^{-3} (value of K_A at 90° K).

(ii) For measurements down to 55° K (triple point of oxygen) we may find N_A and N_D providing $(N_A - N_D) > 2 \times 10^{13}$ cm^{-3} (value of K_A at 55° K).

(iii) For concentrations less than this it is necessary to make measurements down to 20° K (boiling point of hydrogen).

Similar estimates may be made for other impurities in silicon and also other semiconductors.

Fairly extensive measurements of the Hall coefficient have been made on silicon down to 90° K and these have recently been extended to 20° K using liquid hydrogen. The results of these measurements on various samples will be published at a later date.

ACKNOWLEDGEMENTS

The author wishes to thank his colleagues Dr. F. W. G. Rose and Mr. F. Paton of the BTH Research Laboratory for many interesting discussions whilst preparing this manuscript.

REFERENCES

(1) JAMES, J. A., and RICHARDS, D. H. *Nature* [*London*], **176**, p. 1026 (1955).

(2) HANNAY, N. B., and AHEARN, A. J. *Analyt. Chem.*, **26**, p. 1056 (1954).

(3) DEBYE, P. P., and CONWELL, E. M. *Phys. Rev.*, **93**, p. 693 (1954).

(4) MORIN, F. J., and MAITA, J. P. *Phys. Rev.*, **96**, p. 28 (1954).

(5) DRESSELHAUS, G., KIP, A. F., and KITTEL, C. *Phys. Rev.*, **98**, p. 368 (1955).

(6) BURSTEIN, E., PICUS, G., HENVIS, B., and WALLIS, R. *J. Phys. Chem. Solids*, **1**, p. 65 (1956).

(7) DEXTER, R. N., and LAX, B. *Phys. Rev.*, **96**, p. 223 (1954).

NOTES AND NEWS

Correspondence

Action and viscous flow

T. V. Starkey has recently published four papers[1-4] which propound and apply a theory of viscous flow based on the concept of "action." In the first of these[1] he uses the fact that "viscosity" has the same dimensions as "action per unit volume" as a basis for a deduction that solute particles will behave as "macromolecular vortices." From this he deduces three equations for the viscosity of solutions of particles of different shapes, and shows that they are equivalent to the equations of Einstein, Staudinger and Hever, and Huggins. By applying the theory of least action he deduces that particles of a solution (or presumably of a suspension) flowing along a tube, will move towards the tube axis, forming a "Kármán vortex street." From this he deduces a viscosity-concentration relation for high concentrations. He concludes by considering the qualitative result of particle shape upon the viscosity and the change of viscosity with rate of shear, which this theory predicts. The second and third papers[2, 3] consist principally of experimental demonstrations of the fact that suspended particles tend to move to regions of least shear, a phenomenon which he claims can be deduced from the principle of least action, and which he attributes to the Magnus effect.[2] In the most recent paper[4] the theory of least action is shown to lead to the correct velocity profile, both in a circular tube and in a rotating co-axial cylinder viscometer. It is also pointed out that a suspension in viscous flow possesses several features in common with a fluid in turbulent flow.

Certain points in the theory as presented by Dr. Starkey appear to require clarification. Firstly, it does not necessarily follow, from the fact that two quantities have the same dimensional form, that they are physically identifiable. Energy and turning moment both have the dimensions ML^2T^{-2}. It would then not seem to follow from the dimensional similarity of "action per unit volume" and "viscosity" that the two are related, and more justification than has been given by Dr. Starkey is required before the two can be equated with confidence. Secondly, it is questionable whether the principle of least action may be applied without modification to a system in which energy is being converted into heat. A paragraph by Jeans[5] is quoted as justification for such procedure, but an examination of this paragraph shows that the principle of least action must be modified before it is applicable to such a system. Jeans shows that this may be done by the addition of a series of "generalized forces." These are found by subtracting from the externally applied forces those forces caused by the non-reversible processes. Thus the effect of the latter must be known before it is allowable to minimize the action. This is clearly illustrated in the example which Jeans gives (p. 496, Ref. 5). It follows that this reference cannot be used to justify the application of the principle of least action in the way in which it has been done by Dr. Starkey.

The same objection can be levelled against the argument used in the fourth paper[4] which is used to show that the velocity profile of a fluid flowing at low Reynolds' numbers will be that which would dissipate the least energy. In the evaluation of the term δW, which is the work required to displace the system from one configuration to some slightly different configuration, all forces must be included. The force exerted by the fluid because it resists deformation as well as the external force which causes the fluid to move must be considered; and as these are equal and opposite δW must always be zero. Thus the strict application of the principle to conditions where both kinetic and potential energies are absent can produce no prediction.

A further point which would appear to require clarification is the effect of Reynolds' number on the formation of a "Kármán vortex street" along the axis of a tube. The original derivation of the idea that such a configuration would be set up appears to apply to very low Reynolds' numbers, as only viscous forces need be considered. This is also implied when the concept is used to obtain a viscosity-